Sinan Bilir

ARTIFICIAL INTELLIGENCE & DATA SCIENTIST | MLOps, NLP, Generative AI & Cloud Automation

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SUMMERY

I am an AI and Data Scientist experienced in applied machine learning, NLP, generative AI, and business intelligence. I build end-to-end solutions spanning data preparation, model development, SHAP-based explainability, and MLOps deployment. My recent work includes NeoJan, a neonatal risk platform integrating multi-omics and synthetic clinical data, alongside commercial projects in customer behaviour and forecasting. I leverage Azure, cloud workflows, and Docker to deploy reliable, scalable, and user-focused machine learning systems across research and commercial environments.

TECHNICAL SKILLS

Programming, Core Tools & Languages	Python | SQL | Bash | R | Git | GitHub | GitLab | Terraform | Agile | LaTeX | **Turkish:** Native | **English:** Professionally Fluent (C2)
Machine Learning & AI Architectures	Scikit-learn | XGBoost | LightGBM | TensorFlow | Keras | PyTorch | CNNs | RNNs | LSTMs | GANs | Autoencoders | Transfer Learning | SHAP | LIME | Reinforcement Learning | Federated Learning
Natural Language Processing (NLP)	SpaCy | NLTK | Hugging Face Transformers | BERT | GPT-based architectures | Text Classification | Semantic Similarity | Topic Modelling | Information Extraction Pipelines | Prompt Engineering
Cloud, MLOps & Deployment	Azure ML | AWS (S3, EC2, Lambda, SageMaker) | GCP (BigQuery, Vertex AI) | Docker | Kubernetes | MLflow | Airflow | TensorFlow Serving | CI/CD (GitHub Actions, Jenkins) | Model Versioning | Monitoring | Retraining Pipelines | Flask | FastAPI | REST APIs
Data Engineering, Big Data, Sales Engineering	Apache Spark | Hadoop | Databricks | MySQL | PostgreSQL | MongoDB | Feature Engineering | Orchestration | Genomics-oriented Data Processing | Multi-omics Integration | Synthetic Clinical Data | Sales Engineering
Analytics, Visualization & BI	Matplotlib | Seaborn | Plotly | Power BI | Tableau | Looker | Streamlit | Dash | A/B Testing | Statistical Analysis | Business Analysis | Business Intelligence
Evaluation & Performance Metrics	ROC AUC | PR AUC | Precision-Recall | F1 Score | Brier Score | Calibration Analysis | Model Explainability | Cross-Validation Strategies

WORK HISTORY

Machine Learning & AI Systems Multi-Omics Integration Lead Researcher | University of Liverpool

12/2024 – 03/2026 | Remote

- ⇒ Lead an interdisciplinary research initiative spanning machine learning, healthcare AI, and biomedical data science, with direct relevance to regulated, risk-sensitive decision-support environments.
- ⇒ Designed and implemented an end-to-end data and machine le∂arning pipeline integrating gene expression datasets (GSE122340, GSE46995), metabolomics repositories (MetaboLights, Metabolomics Workbench), and synthetic electronic health records from Synthea.
- ⇒ Developed and evaluated Random Forest, XGBoost, and deep neural network models, producing high-recall and calibrated probabilistic outputs for neonatal risk prediction and clinical decision support.
- ⇒ Applied advanced feature engineering, PCA, clustering, and correlation analysis to support biomarker discovery, dimensionality reduction, and cross-domain insight generation.
- ⇒ Built and deployed the NeoJan Streamlit application with structured CSV/JSON ingestion, interactive 3D PCA visualisation, dynamic threshold tuning, and automated clinical-style reporting.
- ⇒ Integrated SHAP-based explainability to improve transparency, auditability, and clinician interpretability of model predictions.
- ⇒ Implemented Dockerised reproducible environments, MLflow experiment tracking, and CI/CD workflows to support scalable and maintainable research operations.
- ⇒ Collaborated within ethics-approved frameworks involving NHS digital health stakeholders and external healthcare research partners, with ongoing preparation for pilot deployment in NHS partner settings.

Data Scientist (Contract) | MGL Licensing

02/2024 – 05/2024 | Remote

- ⇒ Conducted customer segmentation using K-Means clustering to identify behavioural patterns and support retention-focused interventions.
- ⇒ Designed and delivered interactive analytical dashboards to support operational monitoring and strategic decision-making.
- ⇒ Built near real-time analytics workflows to surface behavioural trends, performance bottlenecks, and commercial opportunities.
- ⇒ Applied A/B testing methodologies to optimise engagement strategies and improve return on investment.

Data Scientist (Contractor – Job Simulation) | British Airways (Internship / Job Simulation)

03/2024 – 05/2024 | Remote

- ⇒ Developed a Random Forest classifier to predict customer booking behaviour, improving understanding of conversion drivers and likely booking outcomes.
- ⇒ Performed exploratory data analysis and engineered high-impact features to strengthen model performance.
- ⇒ Communicated analytical findings and business recommendations clearly for non-technical stakeholders.

Lead Sales Operations Business Analyst (Consultant) | Antal TECH Jobs

08/2022 – 04/2023 | Greater London, UK

- ⇒ Built and applied machine learning models to predict job-seeker engagement, contributing to a 20% increase in applications.
- ⇒ Designed automated dashboards using Power BI and SQL to improve recruitment reporting and decision-making.
- ⇒ Used Google Analytics and A/B testing to optimise marketing strategy, contributing to a 15% improvement in conversion rates.

Business Intelligence Consultant | SB Sales and Marketing Consultancy Ltd

04/2019 – 03/2022 | Greater London, UK

- ⇒ Designed and managed business intelligence solutions delivering actionable insight into sales performance and market trends.
- ⇒ Applied predictive modelling in Python to support forecasting, scenario analysis, and business strategy optimisation.
- ⇒ Wrote complex SQL queries to extract, transform, and analyse key metrics for commercial planning and performance review.

Senior Sales Engineer & Data Analyst (EMEA)| Ant Kalip M&G Group

02/2017 – 02/2019 | Kocaeli, Turkiye

- ⇒ Analysed sales trends, KPIs, and operational performance using SAP ERP and Excel, contributing to a 15% increase in sales efficiency.
- ⇒ Used Google Analytics and customer insight data to monitor market performance, support targeted campaigns, and refine product positioning for domestic and international sales activity.
- ⇒ Created Power BI dashboards and Excel reports to improve visibility of commercial and operational metrics and support data-driven decisions.

PROJECTS

NeoJanMulti-Omics Machine Learning for Neonatal Jaundice Prediction Project | University of Liverpool (2025–2026):

Developed the world's first integrated multi-omics machine-learning framework for neonatal jaundice risk prediction. Unified omics + clinical data using robust preprocessing and harmonisation (SCANPY, MetaboAnalystR, biomaRt). Constructed and compared Random Forest, XGBoost, and Deep Neural Network models, achieving PR AUC = 1.0. Embedded explainable AI modules (SHAP/LIME) for transparency and clinician trust. Created a clinician-oriented Streamlit web application allowing real-time patient assessment, threshold visualisation, and decision-support aligned with NHS guidelines. Deployed via Docker + MLflow, ensuring reproducibility, security, and CI/CD integration. Technologies: Python, Scikit-Learn, TensorFlow, Pandas, Streamlit, Plotly, Docker, MLflow, GitHub. Collaborators: NHS Digital Health | University of Liverpool | PhysioNet.

Pneumonia Detection Using Convolutional Neural Networks (CNNs) | University of Liverpool | MSc Data Science & Artificial Intelligence

Developed and deployed a CNN-based deep learning model for pneumonia detection using chest X-ray images, leveraging TensorFlow and Keras. Implemented data preprocessing (normalization, augmentation) and addressed class imbalance using class weight adjustment techniques. Applied Transfer Learning (ResNet15) and K-Fold Cross-Validation, improving model generalization and achieving 97.6% accuracy & 98% recall. Integrated model interpretability techniques (Grad-CAM, SHAP) to enhance explainability for healthcare professionals.

Deep Learning Fashion MNIST Classification Project | University of Liverpool | MSc Data Science & Artificial Intelligence

This project presents the implementation and evaluation of neural network models for image classification on the Fashion MNIST dataset. The models include a Sigmoid activation model as a baseline, a ReLU activation model designed for faster convergence, and a dropout regularisation model developed to reduce overfitting. Each model was trained and evaluated using accuracy and loss metrics.

Predicting Atmospheric Emissions Using Machine Learning Models | University of Liverpool | MSc Data Science & Artificial Intelligence

Developed predictive models to estimate nitrogen oxides (NOx) and particulate matter (PM2.5) emissions across London. Utilized geospatial data from the London Atmospheric Emissions Inventory (LAEI 2019) and applied machine learning algorithms, including Random Forest and XGBoost. Implemented feature engineering techniques such as Principal Component Analysis (PCA) for dimensionality reduction. Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, GeoPandas, Jupyter Notebook

Data Quality Analysis and Cleansing of SIS Faculty Dataset | University of Liverpool | MSc Data Science & Artificial Intelligence

Conducted data quality analysis and cleansing on a Student Information System (SIS) faculty dataset. Addressed issues such as missing values, inconsistent formatting, and redundant data to prepare the dataset for machine learning applications. Technologies: Python, Pandas, Jupyter Notebook.

Secure Client-Server Communication System | University of Liverpool | MSc Data Science & Artificial Intelligence

Developed a secure client-server communication system in Python, emphasizing secure data exchange using symmetric encryption. Supported multiple data serialization formats, including JSON, Binary (Pickle), and XML. Designed with modularity and extensibility in mind. Technologies: Python, Fernet (symmetric encryption), JSON, Pickle, XML

Shortest Path Algorithm Implementation | University of Liverpool | MSc Data Science & Artificial Intelligence

Implemented and tested Floyd's Algorithm for finding the shortest paths in a graph, focusing on understanding algorithmic efficiency. **Technologies**: Python.

EDUCATION

MASTER OF SCIENCE (MSC) | DATA SCIENCE & ARTIFICIAL INTELLIGENCE | UNIVERSITY OF LIVERPOOL | 10/2023 – 04/2026 | **First Class Grade**

Relevant study: Machine Learning in Practice, Deep Learning, Software Development in Practice, Databases and IT Systems, Mathematics and Statistics for AI & Data Science, Cloud Computing; **Dissertation:** Development of a Machine Learning Model &Software Platform for Multi-Omics-Based Prediction of Neonatal Jaundice Risk.

BACHELOR OF SCIENCE (BSC) | INDUSTRIAL ENGINEERING | DOKUZ EYLUL UNIVERSITY | 09/2011 – 08/2016 | **Upper Second ClassGrade**

Relevant study: Operations Research, Production Systems, Statistics, Manufacturing Processes, Quality Planning and Control, Simulation, Optimisation, Logistics, Supply Chain Management, Project Management, Information Systems; **Thesis:** Transition to Just-in-Time Production and Kanban Board Implementation in an Automotive Supplier Firm. Leadership: Regional Manager, ESTIEM, leading international projects in business process optimisation and supply chain management.

RESEARCH & PUBLICATIONS

I. Bilir S., Garron N., Kelm K. (2025). "Development of a Machine Learning Model and Software Platform for Multi-Omics-Based Prediction of Neonatal Jaundice Risk." University of Liverpool — Target Journal: Frontiers in AI – Healthcare Informatics (2026).

II. Bilir S. (2016). "Transition to Just-in-Time Production and Kanban Board Implementation in an Automotive Supplier Firm." BSc Dissertation, Dokuz Eylul University.

CERTIFICATIONS

- **Artificial Intelligence Programme University of Oxford |** To Be Completed
- **Machine Learning with Scikit-Learn LinkedIn |** Sep 2024 *Certificate ID:* 140b356a3304daa311b1459c038c30839a98fd27f86893fcc9ab5c676291f66ef7
- **Business Analysis Foundations | IIBA |** Aug 2024 *Credential ID:* 2d9f9eeb1eb5ae811394e4cba36349b7f912d9ece54ca86e3dd55039d6556250
- **Business Analysis: Essential Tools and Techniques** | Project Management Institute ***ID:***859b5a67274a0baa74d55c36c461f86faeb7a9814204a141935c87376aa307b6
- **Machine Learning with Python: Foundations | LinkedIn** | Jan 2024 *Certificate ID:* 161eefec3e3e4876f725ad25df13df7647f46099eafb6ae6a8ff04713a4872dc1